The dual function for the problem (5.70) is

$$\begin{aligned} g(\nu) &= \inf_{l \preceq x \preceq u} \left(c^T x + \nu^T (Ax - b)\right) \\ &= -b^T \nu - u^T (A^T \nu + c)^- + l^T (A^T \nu + c)^+ \end{aligned}$$

where $y_i^+ = \max\{y_i, 0\}$, $y_i^- = \max\{-y_i, 0\}$. So here we are able to derive an analytical formula for g, which is a concave piecewise-linear function.

The dual problem is the unconstrained problem

$$\text{maximize} \quad -b^T \nu - u^T (A^T \nu + c)^- + l^T (A^T \nu + c)^+, \tag{5.71}$$

which has a quite different form from the dual of the original problem.

(The problems (5.69) and (5.71) are closely related, in fact, equivalent; see exercise 5.8.)

5.8 Theorems of alternatives

5.8.1 Weak alternatives via the dual function

In this section we apply Lagrange duality theory to the problem of determining feasibility of a system of inequalities and equalities

$$f_i(x) \leq 0, \quad i = 1, \ldots, m, \qquad h_i(x) = 0, \quad i = 1, \ldots, p. \tag{5.72}$$

We assume the domain of the inequality system (5.72), $\mathcal{D} = \bigcap_{i=1}^m \mathbf{dom}\, f_i \cap \bigcap_{i=1}^p \mathbf{dom}\, h_i$, is nonempty. We can think of (5.72) as the standard problem (5.1), with objective $f_0 = 0$, *i.e.*,

$$\begin{array}{ll} \text{minimize} & 0 \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p. \end{array} \tag{5.73}$$

This problem has optimal value

$$p^\star = \begin{cases} 0 & \text{(5.72) is feasible} \\ \infty & \text{(5.72) is infeasible,} \end{cases} \tag{5.74}$$

so solving the optimization problem (5.73) is the same as solving the inequality system (5.72).

The dual function

We associate with the inequality system (5.72) the dual function

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} \left(\sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^p \nu_i h_i(x) \right),$$

which is the same as the dual function for the optimization problem (5.73). Since $f_0 = 0$, the dual function is positive homogeneous in (λ, ν): For $\alpha > 0$, $g(\alpha\lambda, \alpha\nu) = \alpha g(\lambda, \nu)$. The dual problem associated with (5.73) is to maximize $g(\lambda, \nu)$ subject to $\lambda \succeq 0$. Since g is homogeneous, the optimal value of this dual problem is given by

$$d^\star = \begin{cases} \infty & \lambda \succeq 0,\ g(\lambda, \nu) > 0 \text{ is feasible} \\ 0 & \lambda \succeq 0,\ g(\lambda, \nu) > 0 \text{ is infeasible.} \end{cases} \tag{5.75}$$

Weak duality tells us that $d^\star \leq p^\star$. Combining this fact with (5.74) and (5.75) yields the following: If the inequality system

$$\lambda \succeq 0, \qquad g(\lambda, \nu) > 0 \tag{5.76}$$

is feasible (which means $d^\star = \infty$), then the inequality system (5.72) is infeasible (since we then have $p^\star = \infty$). Indeed, we can interpret any solution (λ, ν) of the inequalities (5.76) as a *proof* or *certificate* of infeasibility of the system (5.72).

We can restate this implication in terms of feasibility of the original system: If the original inequality system (5.72) is feasible, then the inequality system (5.76) must be infeasible. We can interpret an x which satisfies (5.72) as a certificate establishing infeasibility of the inequality system (5.76).

Two systems of inequalities (and equalities) are called *weak alternatives* if at most one of the two is feasible. Thus, the systems (5.72) and (5.76) are weak alternatives. This is true whether or not the inequalities (5.72) are convex (*i.e.*, f_i convex, h_i affine); moreover, the alternative inequality system (5.76) is always convex (*i.e.*, g is concave and the constraints $\lambda_i \geq 0$ are convex).

Strict inequalities

We can also study feasibility of the *strict* inequality system

$$f_i(x) < 0, \quad i = 1, \ldots, m, \qquad h_i(x) = 0, \quad i = 1, \ldots, p. \tag{5.77}$$

With g defined as for the nonstrict inequality system, we have the alternative inequality system

$$\lambda \succeq 0, \qquad \lambda \neq 0, \qquad g(\lambda, \nu) \geq 0. \tag{5.78}$$

We can show directly that (5.77) and (5.78) are weak alternatives. Suppose there exists an $\tilde{x}$ with $f_i(\tilde{x}) < 0$, $h_i(\tilde{x}) = 0$. Then for any $\lambda \succeq 0$, $\lambda \neq 0$, and ν,

$$\lambda_1 f_1(\tilde{x}) + \cdots + \lambda_m f_m(\tilde{x}) + \nu_1 h_1(\tilde{x}) + \cdots + \nu_p h_p(\tilde{x}) < 0.$$

It follows that

$$\begin{aligned} g(\lambda, \nu) &= \inf_{x \in \mathcal{D}} \left(\sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^p \nu_i h_i(x) \right) \\ &\leq \sum_{i=1}^m \lambda_i f_i(\tilde{x}) + \sum_{i=1}^p \nu_i h_i(\tilde{x}) \\ &< 0. \end{aligned}$$

Therefore, feasibility of (5.77) implies that there does not exist (λ, ν) satisfying (5.78).

Thus, we can prove infeasibility of (5.77) by producing a solution of the system (5.78); we can prove infeasibility of (5.78) by producing a solution of the system (5.77).

5.8.2 Strong alternatives

When the original inequality system is convex, *i.e.*, f_i are convex and h_i are affine, and some type of constraint qualification holds, then the pairs of weak alternatives described above are *strong alternatives*, which means that *exactly one* of the two alternatives holds. In other words, each of the inequality systems is feasible if and only if the other is infeasible.

In this section we assume that f_i are convex and h_i are affine, so the inequality system (5.72) can be expressed as

$$f_i(x) \leq 0, \quad i = 1, \ldots, m, \qquad Ax = b,$$

where $A \in \mathbf{R}^{p \times n}$.

Strict inequalities

We first study the strict inequality system

$$f_i(x) < 0, \quad i = 1, \ldots, m, \qquad Ax = b, \tag{5.79}$$

and its alternative

$$\lambda \succeq 0, \qquad \lambda \neq 0, \qquad g(\lambda, \nu) \geq 0. \tag{5.80}$$

We need one technical condition: There exists an $x \in \mathbf{relint}\, \mathcal{D}$ with $Ax = b$. In other words we not only assume that the linear equality constraints are consistent, but also that they have a solution in $\mathbf{relint}\, \mathcal{D}$. (Very often $\mathcal{D} = \mathbf{R}^n$, so the condition is satisfied if the equality constraints are consistent.) Under this condition, exactly one of the inequality systems (5.79) and (5.80) is feasible. In other words, the inequality systems (5.79) and (5.80) are strong alternatives.

We will establish this result by considering the related optimization problem

$$\begin{array}{ll} \text{minimize} & s \\ \text{subject to} & f_i(x) - s \leq 0, \quad i = 1, \ldots, m \\ & Ax = b \end{array} \tag{5.81}$$

with variables x, s, and domain $\mathcal{D} \times \mathbf{R}$. The optimal value $p^\star$ of this problem is negative if and only if there exists a solution to the strict inequality system (5.79).

The Lagrange dual function for the problem (5.81) is

$$\inf_{x \in \mathcal{D},\, s} \left(s + \sum_{i=1}^{m} \lambda_i (f_i(x) - s) + \nu^T (Ax - b) \right) = \begin{cases} g(\lambda, \nu) & \mathbf{1}^T \lambda = 1 \\ -\infty & \text{otherwise.} \end{cases}$$

Therefore we can express the dual problem of (5.81) as

$$\begin{array}{ll} \text{maximize} & g(\lambda,\nu) \\ \text{subject to} & \lambda \succeq 0, \quad \mathbf{1}^T\lambda = 1. \end{array}$$

Now we observe that Slater's condition holds for the problem (5.81). By the hypothesis there exists an $\tilde{x} \in \mathbf{relint}\,\mathcal{D}$ with $A\tilde{x} = b$. Choosing any $\tilde{s} > \max_i f_i(\tilde{x})$ yields a point $(\tilde{x}, \tilde{s})$ which is strictly feasible for (5.81). Therefore we have $d^\star = p^\star$, and the dual optimum $d^\star$ is attained. In other words, there exist $(\lambda^\star, \nu^\star)$ such that

$$g(\lambda^\star, \nu^\star) = p^\star, \qquad \lambda^\star \succeq 0, \qquad \mathbf{1}^T\lambda^\star = 1. \tag{5.82}$$

Now suppose that the strict inequality system (5.79) is infeasible, which means that $p^\star \geq 0$. Then $(\lambda^\star, \nu^\star)$ from (5.82) satisfy the alternate inequality system (5.80). Similarly, if the alternate inequality system (5.80) is feasible, then $d^\star = p^\star \geq 0$, which shows that the strict inequality system (5.79) is infeasible. Thus, the inequality systems (5.79) and (5.80) are strong alternatives; each is feasible if and only if the other is not.

Nonstrict inequalities

We now consider the nonstrict inequality system

$$f_i(x) \leq 0, \quad i = 1, \ldots, m, \qquad Ax = b, \tag{5.83}$$

and its alternative

$$\lambda \succeq 0, \qquad g(\lambda, \nu) > 0. \tag{5.84}$$

We will show these are strong alternatives, provided the following conditions hold: There exists an $x \in \mathbf{relint}\,\mathcal{D}$ with $Ax = b$, and the optimal value $p^\star$ of (5.81) is attained. This holds, for example, if $\mathcal{D} = \mathbf{R}^n$ and $\max_i f_i(x) \to \infty$ as $x \to \infty$. With these assumptions we have, as in the strict case, that $p^\star = d^\star$, and that both the primal and dual optimal values are attained. Now suppose that the nonstrict inequality system (5.83) is infeasible, which means that $p^\star > 0$. (Here we use the assumption that the primal optimal value is attained.) Then $(\lambda^\star, \nu^\star)$ from (5.82) satisfy the alternate inequality system (5.84). Thus, the inequality systems (5.83) and (5.84) are strong alternatives; each is feasible if and only if the other is not.

5.8.3 Examples

Linear inequalities

Consider the system of linear inequalities $Ax \preceq b$. The dual function is

$$g(\lambda) = \inf_x \lambda^T(Ax - b) = \begin{cases} -b^T\lambda & A^T\lambda = 0 \\ -\infty & \text{otherwise.} \end{cases}$$

The alternative inequality system is therefore

$$\lambda \succeq 0, \qquad A^T\lambda = 0, \qquad b^T\lambda < 0.$$

These are, in fact, strong alternatives. This follows since the optimum in the related problem (5.81) is achieved, unless it is unbounded below.

We now consider the system of strict linear inequalities $Ax \prec b$, which has the strong alternative system

$$\lambda \succeq 0, \qquad \lambda \neq 0, \qquad A^T\lambda = 0, \qquad b^T\lambda \leq 0.$$

In fact we have encountered (and proved) this result before, in §2.5.1; see (2.17) and (2.18) (on page 50).

Intersection of ellipsoids

We consider m ellipsoids, described as

$$\mathcal{E}_i = \{x \mid f_i(x) \leq 0\},$$

with $f_i(x) = x^TA_ix + 2b_i^Tx + c_i$, $i = 1, \ldots, m$, where $A_i \in \mathbf{S}^n_{++}$. We ask when the intersection of these ellipsoids has nonempty interior. This is equivalent to feasibility of the set of strict quadratic inequalities

$$f_i(x) = x^TA_ix + 2b_i^Tx + c_i < 0, \quad i = 1, \ldots, m. \tag{5.85}$$

The dual function g is

$$\begin{aligned} g(\lambda) &= \inf_x \left(x^TA(\lambda)x + 2b(\lambda)^Tx + c(\lambda)\right) \\ &= \begin{cases} -b(\lambda)^TA(\lambda)^\dagger b(\lambda) + c(\lambda) & A(\lambda) \succeq 0, \quad b(\lambda) \in \mathcal{R}(A(\lambda)) \\ -\infty & \text{otherwise,} \end{cases} \end{aligned}$$

where

$$A(\lambda) = \sum_{i=1}^m \lambda_i A_i, \qquad b(\lambda) = \sum_{i=1}^m \lambda_i b_i, \qquad c(\lambda) = \sum_{i=1}^m \lambda_i c_i.$$

Note that for $\lambda \succeq 0$, $\lambda \neq 0$, we have $A(\lambda) \succ 0$, so we can simplify the expression for the dual function as

$$g(\lambda) = -b(\lambda)^TA(\lambda)^{-1}b(\lambda) + c(\lambda).$$

The strong alternative of the system (5.85) is therefore

$$\lambda \succeq 0, \qquad \lambda \neq 0, \qquad -b(\lambda)^TA(\lambda)^{-1}b(\lambda) + c(\lambda) \geq 0. \tag{5.86}$$

We can give a simple geometric interpretation of this pair of strong alternatives. For any nonzero $\lambda \succeq 0$, the (possibly empty) ellipsoid

$$\mathcal{E}_\lambda = \{x \mid x^TA(\lambda)x + 2b(\lambda)^Tx + c(\lambda) \leq 0\}$$

contains $\mathcal{E}_1 \cap \cdots \cap \mathcal{E}_m$, since $f_i(x) \leq 0$ implies $\sum_{i=1}^m \lambda_i f_i(x) \leq 0$. Now, $\mathcal{E}_\lambda$ has empty interior if and only if

$$\inf_x \left(x^TA(\lambda)x + 2b(\lambda)^Tx + c(\lambda)\right) = -b(\lambda)^TA(\lambda)^{-1}b(\lambda) + c(\lambda) \geq 0.$$

Therefore the alternative system (5.86) means that $\mathcal{E}_\lambda$ has empty interior.

Weak duality is obvious: If (5.86) holds, then $\mathcal{E}_\lambda$ contains the intersection $\mathcal{E}_1 \cap \cdots \cap \mathcal{E}_m$, and has empty interior, so naturally the intersection has empty interior. The fact that these are strong alternatives states the (not obvious) fact that if the intersection $\mathcal{E}_1 \cap \cdots \cap \mathcal{E}_m$ has empty interior, then we can construct an ellipsoid $\mathcal{E}_\lambda$ that contains the intersection and has empty interior.

Farkas' lemma

In this section we describe a pair of strong alternatives for a mixture of strict and nonstrict linear inequalities, known as *Farkas' lemma*: The system of inequalities

$$Ax \preceq 0, \qquad c^T x < 0, \tag{5.87}$$

where $A \in \mathbf{R}^{m\times n}$ and $c \in \mathbf{R}^n$, and the system of equalities and inequalities

$$A^T y + c = 0, \qquad y \succeq 0, \tag{5.88}$$

are strong alternatives.

We can prove Farkas' lemma directly, using LP duality. Consider the LP

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax \preceq 0, \end{array} \tag{5.89}$$

and its dual

$$\begin{array}{ll} \text{maximize} & 0 \\ \text{subject to} & A^T y + c = 0 \\ & y \succeq 0. \end{array} \tag{5.90}$$

The primal LP (5.89) is homogeneous, and so has optimal value 0, if (5.87) is not feasible, and optimal value $-\infty$, if (5.87) is feasible. The dual LP (5.90) has optimal value 0, if (5.88) is feasible, and optimal value $-\infty$, if (5.88) is infeasible.

Since $x = 0$ is feasible in (5.89), we can rule out the one case in which strong duality can fail for LPs, so we must have $p^\star = d^\star$. Combined with the remarks above, this shows that (5.87) and (5.88) are strong alternatives.

Example 5.10 *Arbitrage-free bounds on price.* We consider a set of n assets, with prices at the beginning of an investment period $p_1, \ldots, p_n$, respectively. At the end of the investment period, the value of the assets is $v_1, \ldots, v_n$. If $x_1, \ldots, x_n$ represents the initial investment in each asset (with $x_j < 0$ meaning a short position in asset j), the cost of the initial investment is $p^T x$, and the final value of the investment is $v^T x$.

The value of the assets at the end of the investment period, v, is uncertain. We will assume that only m possible scenarios, or outcomes, are possible. If outcome i occurs, the final value of the assets is $v^{(i)}$, and therefore, the overall value of the investments is $v^{(i)T} x$.

If there is an investment vector x with $p^T x < 0$, and in all possible scenarios, the final value is nonnegative, *i.e.*, $v^{(i)T} x \geq 0$ for $i = 1, \ldots, m$, then an *arbitrage* is said to exist. The condition $p^T x < 0$ means you are *paid* to accept the investment mix, and the condition $v^{(i)T} x \geq 0$ for $i = 1, \ldots, m$ means that no matter what outcome occurs, the final value is nonnegative, so an arbitrage corresponds to a guaranteed money-making investment strategy. It is generally assumed that the prices and values are such that no arbitrage exists. This means that the inequality system

$$Vx \succeq 0, \qquad p^T x < 0$$

is infeasible, where $V_{ij} = v_j^{(i)}$.

Using Farkas' lemma, we have no arbitrage if and only if there exists y such that

$$-V^T y + p = 0, \qquad y \succeq 0.$$